



Semester Project	
Course Title: Development Operations Lab	Course Code: SE202L
Class & Section: BS (CS) – Spring 2026	Instructor: M.Naeem
Project Submission Deadline: One week prior to final exams	Group Members: 3 or less
Online Form Submission Deadline: 05– May – 2026	

Dear Students,

As part of the SE202L – Development Operations Lab, all students are required to implement a complete DevOps lifecycle to solve a real-world problem. The goal of this project is to ensure that students gain hands-on experience in Linux, Git, Docker, CI/CD pipelines, cloud deployment, and infrastructure automation by building a fully functional system.

All students must complete a semester project following the rules and conditions below.

Rules and Conditions

1. Project Registration

- ✓ Each group must:
 - Select a unique project title
 - Mention names and registration numbers of all members
- ✓ A group must not exceed 3 students
- ✓ A student cannot be part of multiple groups
- ✓ Only one member will submit the project details

2. Group & Title Rules

- ✓ Once registered, project title and group cannot be changed
- ✓ No two groups can select the same project
- ✓ Titles will be approved on a first-come, first-served basis

3. Important Features

- ✓ Students are required to develop:
 - A Containerized Web Application with a Complete DevOps Pipeline
 - The system must demonstrate the end-to-end automation of application deployment using DevOps practices.

Each project must include the following core DevOps components:

- ✓ **Version Control (Git & GitHub)**
 - Repository with:
 - Proper commits from all members
 - Branching & merging
 - README documentation
- ✓ **Containerization (Docker)**
 - Create:
 - Dockerfile
 - Docker Image
 - Run application inside container

- Use Docker Compose (if multi-service)
- ✓ **CI/CD Pipeline (GitHub Actions)**
 - Automate:
 - Build process
 - Docker image creation
 - Pipeline should trigger on:
 - Push to main branch
- ✓ **Cloud Deployment (AWS EC2)**
 - Launch EC2 instance
 - Deploy application
- ✓ **Infrastructure as Code (Terraform) (Optional)**
 - Write scripts to:
 - Provision EC2 instance
 - Manage infrastructure automatically

4. Feature to Requirement Breakdown

Each feature must be broken into clear technical requirements.

- ✓ Req 1: Detect code push on GitHub
- ✓ Req 2: Trigger CI/CD pipeline
- ✓ Req 3: Build Docker image
- ✓ Req 4: Push image to registry (optional)
- ✓ Req 5: Connect to AWS EC2
- ✓ Req 6: Deploy container on server
- ✓ Req 7: Expose application on browser

5. Documentation & GitHub

- ✓ Create a complete project report (Word file) including:
 - System architecture diagram
 - Implementation steps
 - Screenshots
- ✓ Upload all code to GitHub repository
 - Contributions from all members required
 - Commits must be distributed over time

6. Deadlines

Project submission: One week before final exams

Late submission will result in penalty or zero marks

7. Marking Criteria:

- ✓ **Documentation & GitHub (20%)**
 - Proper report
 - GitHub activity and commits
- ✓ **Implementation & DevOps Pipeline (40%)**
 - Docker + CI/CD + AWS + Terraform integration
- ✓ **Project Viva & Demonstration (40%)**
 - Working pipeline demonstration
 - Understanding of tools and concepts

